

L1 – Trig Review and Special Angles

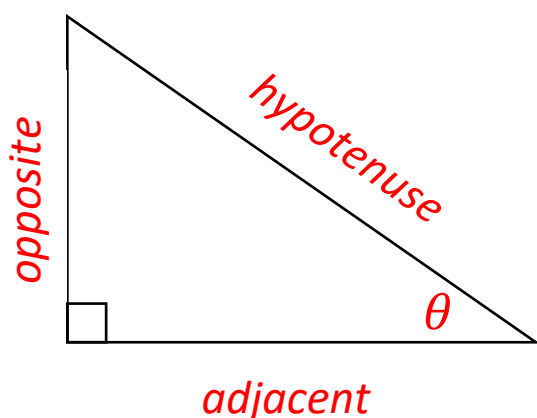
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Part 1: Trig Review

Your main takeaway from grade 10 trigonometry should have been:

If we know a right triangle has an angle of θ , all other right triangles with an angle of θ are **SIMILAR** and therefore have **EQUIVALENT** ratios of corresponding sides.

There are three primary trigonometric ratios for right angled triangles. **Sine**, **Cosine**, and **Tangent**.



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

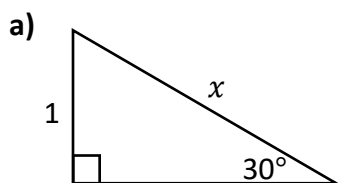
$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Acronym: SOHCAHTOA

S $\frac{O}{H}$ **C** $\frac{A}{H}$ **T** $\frac{O}{A}$

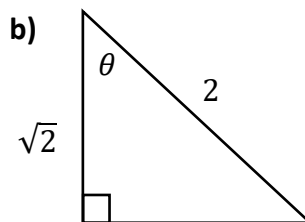
Example 1: Find the indicated missing side or angle of each triangle



$$\sin 30 = \frac{1}{x}$$

$$x = \frac{1}{\sin 30}$$

$$x = 2$$



$$\cos \theta = \frac{\sqrt{2}}{2}$$

$$\theta = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

$$\theta = 45^\circ$$

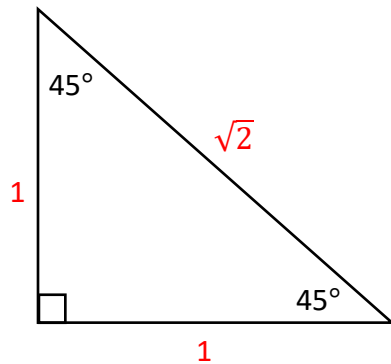
Part 2: Special Angles

There are 2 special triangles:

i) isosceles: $45^\circ - 45^\circ - 90^\circ$

ii) half equilateral: $30^\circ - 60^\circ - 90^\circ$

i)

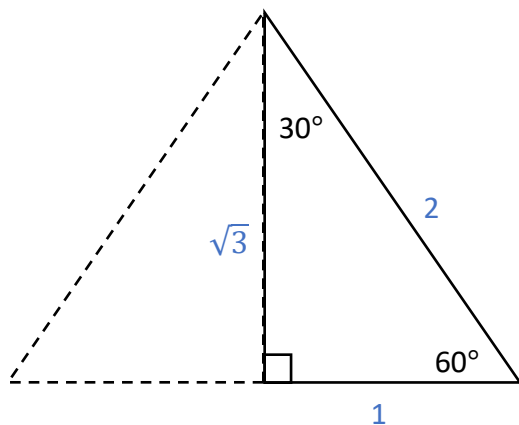


$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{1}{1} = 1$$

ii)



$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

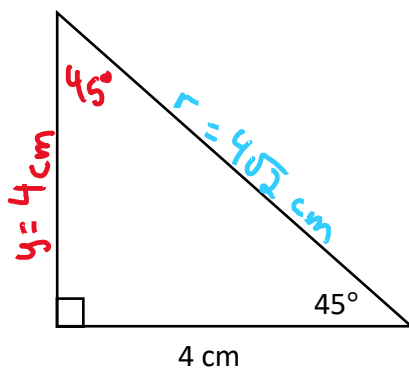
$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 60^\circ = \frac{\sqrt{3}}{1} = \sqrt{3}$$

All sized right triangles with these angles are **SIMILAR** and therefore will have the same ratios of corresponding sides. Therefore, we can use these 2 special triangles to get **EXACT** values for trig ratios involving a 30° , 45° , or 60° reference angle AND we don't need a calculator!

Example 2: Use special triangles to find the EXACT values of all sides and angles

a)



$$\tan 45^\circ = \frac{y}{x}$$

$$1 = \frac{y}{x}$$

$$y = x$$

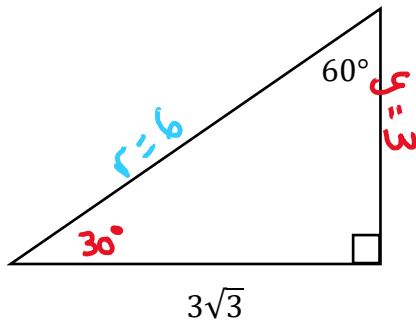
$$\cos 45^\circ = \frac{x}{r}$$

$$\frac{1}{\sqrt{2}} = \frac{x}{r}$$

$$r = x\sqrt{2}$$

6

b)



$$\tan 60 = \frac{3\sqrt{3}}{y}$$

$$\sqrt{3} = \frac{3\sqrt{3}}{y}$$

$$y = \frac{3\sqrt{3}}{\sqrt{3}}$$

$$y = 3$$

$$\sin 60 = \frac{3\sqrt{3}}{r}$$

$$\frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{r}$$

$$r\sqrt{3} = 6\sqrt{3}$$

$$r = \frac{6\sqrt{3}}{\sqrt{3}}$$

$$r = 6$$

Example 3: Determine the exact value of...

a) $(\sin 45^\circ)(\cos 45^\circ) + (\sin 30^\circ)(\sin 60^\circ)$

$$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{4}$$

$$= \frac{2}{4} + \frac{\sqrt{3}}{4}$$

$$= \frac{2+\sqrt{3}}{4}$$

b) $\frac{\sin^2 30^\circ}{1 - \cos 30^\circ}$

$$= \frac{(\sin 30)(\sin 30)}{1 - \cos 30}$$

$$= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)}{1 - \frac{\sqrt{3}}{2}}$$

$$= \frac{\left(\frac{1}{4}\right)}{\left(\frac{2-\sqrt{3}}{2}\right)}$$

$$= \frac{1}{4} \times \frac{2}{2-\sqrt{3}}$$

$$= \frac{1}{2(2-\sqrt{3})}$$

$$= \frac{1}{4-2\sqrt{3}}$$

Part 3: Rationalizing the Denominator

Fractions should be simplified so that the denominator contains only rational numbers.

Example 4: Rationalize the denominator for each of the following expressions

a) $\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$$= \frac{\sqrt{2}}{\sqrt{4}}$$

$$= \frac{\sqrt{2}}{2}$$

b) $\frac{3}{1+\sqrt{5}} \times \frac{1-\sqrt{5}}{1-\sqrt{5}}$

← multiplying by the 'conjugate'

$$= \frac{3(1-\sqrt{5})}{(1)^2 - (\sqrt{5})^2}$$

$$= \frac{3-3\sqrt{5}}{1-5}$$

$$= \frac{3-3\sqrt{5}}{-4}$$

$$= \frac{-3+3\sqrt{5}}{4}$$